



Storage – Speed vs Capacity

Balancing speed, capacity, endurance and cost for modern storage needs

Our digital lives hinge on fast, reliable storage. Whether you're launching a game, transferring gigabytes of footage or training AI models, the drive's ability to move data dictates your PC's responsiveness. In 2025 choices range from affordable SATA SSDs to bleeding-edge PCI Express (PCIe) Gen 5 NVMe drives. Each option involves trade-offs between raw speed, capacity, endurance, heat and cost. This chapter cuts through jargon so you can choose the right drive and keep it running smoothly.

Interfaces and generations

SATA SSDs connect via a legacy interface designed for mechanical disks and cap out around 550 MB/s. They are still useful for upgrading older PCs but are no match for NVMe SSDs, which plug into the PCIe bus through M.2 slots. NVMe drives exploit multiple PCIe lanes to deliver several gigabytes per second of throughput and far lower latency. Unless you're stuck with an old motherboard, NVMe should be your default.

PCIe bandwidth doubles every generation: a Gen 3 NVMe drive can read roughly 3.5 GB/s; Gen 4 doubles that to about 7